



Department of Mathematics & Natural Sciences

MAT 110: Differential Calculus & Coordinate Geometry (Mathematics I)

Summer 2024

“Pure mathematics is, in its way, the poetry of logical ideas” – Albert Einstein

Consultation Schedule

Will be provided by the instructor.

Instructor's Information

Will be provided by the instructor.

Rationale

The study of this course helps to learn how things change. It provides a framework for modeling systems in which there is change, and a way to deduce the predictions of such models.

Course Aims and Outcomes

Content in this course will be adaptive, allowing students to achieve a certain concept before moving on to the next level.

Learning Outcomes

This course is designed to provide science/ engineering students an intense foundational introduction to the fundamental concepts in mathematics. After completing this course a student will be benefitted in the following ways:

- Calculate limits of single and multivariable functions both analytically and graphically.
- Comprehend the notations of differentiability and continuity for single variable functions.
- Calculate the 1st and higher order derivatives of single and multivariable functions, wherever they exist.
- Apply the single variable mean value theorem (for functions of a single variable).
- Find maxima, minima, inflection points and saddle points.
- Apply differentiation to optimization problems in both single and multivariable calculus (Lagrange Multipliers).
- Calculate Taylor Series (Single and Multivariable) and use them to find approximate values of functions.
- Calculate divergence, curl and gradient and interpret them.
- Understand and use spherical and cylindrical coordinates in three dimensions.
- Understand and apply the knowledge of conic sections to real world problems.

Course Contents:

Limits, continuity, differentiation, applications of differentiation, optimization, Taylor and Maclaurin Series, Partial derivatives, optimization of multivariable functions, gradient, curl, divergence and their applications, conic sections, polar, spherical and cylindrical coordinates.

Marks Distribution

Attendance = 5%

Assignment (best 3 out of 4) = 15%

Quiz (best 2 out of 3) = 25%

Midterm = 20%

Final = 35%

Total = 100%

Attendance	Marks
90% and above	5
85% - 89%	4
80% - 84%	3
75% - 79%	2
70% - 74%	1
Less than 70% (Unable to sit for Final Exam)	

Lecture Plan

Week 1	Introduction to Limits and Continuity
Week 2	Introduction to Differentiation and Techniques of Differentiation
Week 3	Leibnitz' product rule, Gradient, Tangent, Linear approximation
Week 4	Maxima, Minima, Optimization problems, Roll's and Mean value theorem
Week 5	Taylor and Maclaurin polynomials for single variable functions
Mid Term	
Week 6	Partial derivatives and Chain rules for partial derivatives
Week 7	Maxima and Minima of multivariable functions and Taylor expansion of multivariable functions
Week 8	Vector Calculus (Fields, Gradient, Divergence and Curl)
Week 9	Lagrange Multipliers
Week 10	Conic sections (parabola, circle, ellipse and hyperbola)
Week 11	Coordinate Geometry (polar, cylindrical, spherical coordinates and conic sections in polar coordinates)
Final	

Administrative information and Course Requirements

- There will be **no makeup quizzes**.
- For **makeup midterm**, a student must *submit application* through the corresponding chair/dean of the department within an appropriate time range.
- No students will be allowed to sit for final exam if he/she misses 30% of the total classes.
- You will find all supporting documents at [Drive link]
- You are strongly encouraged to drop by my office to ask questions and discuss problems.
- Pass mark: 50%

Reference Book:

- Calculus, Early Transcendentals (10th Edition) by **Howard Anton**
- Schum's Outlines Vector Analysis and an introduction to Tensor Analysis by **Murray R. Spiegel**
- Calculus, Early Transcendentals (9th Edition) by **Stewart, Clegg, Watson**

Academic Integrity

Each student in this course is expected to abide by the BRAC University Code of Academic Integrity. Any work submitted by a student in this course for academic credit will be the student's own work. Also refer to plagiarism policy that will be on the web.

You are encouraged to study together and to discuss information and concepts covered in lecture and the sections with other students. You can give "consulting" help to or receive "consulting" help from such students. However, this permissible cooperation should never involve one student having possession of a copy of all or part of work done by someone else, in the form of an e-mail, an e-mail attachment file, a diskette, or a hard copy.

If copying occurs, both the student who copied work from another student and the student who gave material to be copied will both automatically receive a zero for the quizzes and assigned work at home. Penalty for violation of this Code can also be extended to include failure of the course and University disciplinary action.

During examinations, you must do your own work. Talking or discussion is not permitted during the examinations, nor May you compare papers, copy from others, or collaborate in any way. Any collaborative behavior during the examinations will result in failure of the exam, and may lead to failure of the course and University disciplinary action.

Attendance Requirement

Attendance in a class is mandatory.

A student with class attendance below 70% must consult the instructor in order to discuss the completion of the course.

Student absent in 3 consecutive classes will require a Chairperson's permission to attend the following classes.

Student unable to attend classes for known reason, must apply to the Chairperson, with copies to the course teacher and the Registrar's Office, mentioning the dates and reasons for absence. If it is not possible to inform it in advance, the Chairperson must be informed by sending an application through messenger, post, fax, or email.

Source: <<https://www.bracu.ac.bd/academics/policies-and-procedures>>



MAT 120: Integral Calculus and Differential Equations

A. Course General Information:

Course Code:	MAT120
Course Title:	Integral Calculus and Differential Equations
Credit Hours:	3
Category	Offered by MNS for other programs
Type:	Theory + Laboratory
Prerequisites:	MAT 110

B. Course Catalog Description (Content):

Integral Calculus:

Integration using Riemann Sums, Integration and Antiderivatives, Indefinite & Definite integrals, Integration by Substitution, Improper Integrals, Integration by parts, Reduction formula, Integration by Trigonometric Substitution, Integration of Rational Function, Beta and Gamma Function, Length of a curve, Surface area of revolution, Basic concept of Double integral, Double integral in Cartesian and Polar Coordinate system, Concept of Triple Integral, Triple integral in Cartesian, Cylindrical and Spherical Coordinate system, Change of variables: Jacobian.

Differential Equations:

Introduction to Differential Equation, Separable Variables, Linear Equations, Integrating Factor, Exact Equation, A non-Exact DE made Exact, Higher order DE: homogeneous and nonhomogeneous cases, Modeling with 1st order ODE.

C. Rationale of the Course: Integral calculus and Differential Equations assists as a foundation for the higher branch of mathematics which can help us to solve various types of real-world problems. The aim of this course is to build the most basic foundation of integral calculus for undergraduate engineering students. It is considered one of the most important courses for engineering majors for it lays the foundation of the key concepts of differential equations.

D. Course Objective:

The goal of this course:

- Clear understanding of differential and integral calculus.
- Proving mathematical results in an organized way.
- Apply the results to solve interesting problems efficiently.
- Enhance critical thinking ability as an engineer: understanding the importance of integral calculus differential equations in their respective fields.

E. Course Outcomes (COs):

Upon successful completion of this course, students will be able to

SL.	CO Description
CO1	Comprehend integration for single and multivariable functions
CO2	Generate model in several branches of engineering using integral calculus
CO3	Understand the theory of differential equations
CO4	Apply the knowledge of differential equations to solve real world problems

F. Course plan specifying content, COs, co-curricular activities (if any), teaching learning and assessment strategy mapped with COs:

Class Week	Topic	Teaching-Learning Strategy	Assessment Strategy	Corresponding COs
1	Definition of integration, Antiderivatives.	Class lecture, Class discussion, Consultations	Assignment, Quiz, Examinations	CO1, CO2
2	Techniques of integration for indefinite integrals and definite integrals (u-substitution, trigonometric substitution, integration by parts, reduction formula, integration of rational function).	Class lecture, Class discussion, Consultations	Assignment, Quiz, Examinations	CO1, CO2
3	Improper integrals, arc lengths, surface area of revolution.	Class lecture, Class discussion, Consultations	Assignment, Quiz, Examinations	CO1, CO2
4	Gamma and Beta Function	Class lecture, Class discussion, Consultations	Assignment, Quiz, Examinations	CO1, CO2
****The topics above could be the tentative syllabus of midterm exam ****				

5	Double integrals: rectangular and polar coordinates	Class lecture, Class discussion, Consultations	Assignment, Quiz, Examinations	CO1, CO2
6	Triple integrals: rectangular, cylindrical and spherical coordinates	Class lecture, Class discussion, Consultations	Assignment, Quiz, Examinations	CO1, CO2
7	Change of variables: Jacobian	Class lecture, Class discussion, Consultations	Assignment, Quiz, Examinations	CO1, CO2
8	Introduction to Differential Equations	Class lecture, Class discussion, Consultations	Assignment, Quiz, Examinations	CO3, CO4
9	First Order Ordinary Differential Equations: Separable and Linear ODE.	Class lecture, Class discussion, Consultations	Assignment, Quiz, Examinations	CO3, CO4
10	Exact, Non-Exact ODE	Class lecture, Class discussion, Consultations	Assignment, Quiz, Examinations	CO3, CO4
11	Higher Order Ordinary Differential Equations: Homogeneous and Non-Homogeneous ODE	Class lecture, Class discussion, Consultations	Assignment, Quiz, Examinations	CO3, CO4
12	Modelling with 1st order ODE.	Class lecture, Class discussion, Consultations	Assignment, Quiz, Examinations	CO3, CO4
**** The topics above could be the tentative syllabus of final exam ****				

G. Learning Materials:

References:

1. Calculus, Early Transcendentals (10th Edition) by Howard Anton.
2. A First course in Differential Equations with Modeling Applications (7th edition) by Dennis G. Zill.

H. Delivery method: It will mainly involve two lectures of 1 hour and 20 minutes durations every week. Different teaching aids e.g. power point presentation will be used when it will be deemed necessary. Class room discussion and problem-solving sessions will supplement the lecture presentations.

I. Assessment and Evaluation:

Assessment Strategy

Evaluation of the students will be done on the basis of the mark's distribution scheme. The mid-term examination will be held around the middle of the semester. The quizzes and assignments are based on the lectures and there will be no make-up for missed quizzes. During quizzes, mid-term and final exams each student must work alone.

Marks distribution: The total mark for this course is 100. The marks will be allocated as follows:

Attendance	Quiz	Lab	Assignment	Midterm	Final
5%	15%	20%	10%	20%	30%

Attendance policy: Arriving 15 minutes late or more is automatically absent. If the total attendance (both lecture and lab classes) is less than 70% of the total number of classes, the student will not be allowed to appear in the mid-semester and semester final exams irrespective of the reason of absences, and he/she will be listed as an absentee. 5% of the total marks allocated for class attendance will be awarded according to the following scheme.

Attendance	Marks
90% and above	5
85% - 89%	4
80% - 84%	3
75% - 79%	2
70% - 74%	1
Less than 70%	0(Absentee)

Make-up Procedures: Make-up examinations will only be allowed to the students with extreme medical conditions or death in the immediate family during the semester. In such cases, permission to sit for the make-up exam is to be recommended by the course teacher and approved by the departmental head offering the course. The candidate must pay a make-up exam fee as per the university rule.

- J. Grading policy:** The grades and grade points will be calculated as per the Brac University grading system which is as follows:

% of marks obtained	Letter grade	Grade sign	Grade points
97 - 100	A ⁺	plus	4.00
90 - < 97	A	plain	4.00
85 - < 90	A ⁻	minus	3.70
80 - < 85	B ⁺	plus	3.30
75 - < 80	B	plain	3.00
70 - < 75	B ⁻	minus	2.70
65 - < 70	C ⁺	plus	2.30
60 - < 65	C	plain	2.00
57 - < 60	C -	minus	1.70
55 - < 57	D ⁺	plus	1.30
52 - <55	D	plain	1.00
50 - <52	D-	minus	0.70
< 50	F	plain	0.00



Semester: Fall 2025

Course No: MAT 216 (Section: 03)

Course Title: Linear Algebra & Fourier Analysis (Mathematics IV)

Credit Hours: 3.0

Instructor Information

Dr. Sharmina Hussain (SHH)

Associate Professor

Department of Mathematics and Natural Sciences (MNS)

Office: 4G50 E-mail: sharmina@bracu.ac.bd

Background:

- Ph.D. in Applied Mathematics, University of Dhaka, Bangladesh
- M.Sc. in Applied Mathematics, University of Dhaka, Bangladesh
- M.Eng. in Mechanical Engineering, University of Victoria, Canada
- B.Sc. (Hons) in Mathematics, University of Dhaka, Bangladesh

Class Schedule and Counseling Hours

Course ID	Class Schedule	Counseling Hour
MAT 216	Sunday, Tuesday: 8:00 am – 9:20 am (Section 7)	Thursday 12:00 pm – 3:00 pm Room: 4G50
	Monday, Wednesday: 8:00 am – 9:20 am (Section 13)	
	Monday, Wednesday: 9:30 am – 10:50 am (Section 14)	Sunday, Monday, Tuesday: 12:00 noon – 1:00 pm Room: 4G50

Course Objective

Throughout this course, it is expected that students will be able to do the following:

- Construct, or give examples of, mathematical expressions that involve vectors, matrices, and linear systems of linear equations.
- Evaluate mathematical expressions to compute quantities that deal with linear systems and eigenvalue problems.
- Analyze mathematical statements and expressions (for example, to assess whether a particular statement is accurate, or to describe solutions of systems in terms of existence and uniqueness).
- Write logical progressions of precise mathematical statements to justify and communicate reasoning.

- Apply linear algebra concepts to model, solve, and analyze real-world situations.
- Find the Fourier series representation of a function of one variable.

These topics are of great importance to the electrical engineering and physics communities.

Course Prerequisite: MAT 110 and MAT 120.

Recommended Books

- *Elementary Linear Algebra*, 9th edition
Howard Anton and Chris Rorres
Copyright © 2005, John Wiley & Sons, Inc.
ISBN: 0-471-43329-2
- *Calculus*, 10th edition
Howard Anton, Irl Bivens and Stephen Davis
Copyright © 2012, Anton Textbooks, Inc., John Wiley & Sons, Inc.
ISBN: 978-0-470-64769-1
- *Theory and Problems of Fourier Analysis with Applications to Boundary Value Problems*
Murray R. Spiegel
Schaum's Outline Series
Copyright © 1974 by McGraw-Hill, Inc.
ISBN: 0-07-060219-0

Academic Integrity

Each student in this course is expected to abide by the BRAC University Code of Academic Integrity. Any work submitted by a student in this course for academic credit will be the student's own work. Also refer to the plagiarism policy available on the web.

You are encouraged to study together and to discuss information and concepts covered in lecture with other students. You can give "consulting" help to or receive "consulting" help from such students. However, this permissible cooperation should never involve one student having possession of a copy of all or part of work done by someone else, in the form of an e-mail, an e-mail attachment file, a diskette, or a hard copy.

If copying occurs, both the student who copied work from another student and the student who gave material to be copied will automatically receive a zero for the quizzes and assigned work. Penalty for violation of this Code can also be extended to include failure of the course and University disciplinary action.

During examinations, you must do your own work. Talking or discussion is not permitted during examinations, nor may you compare papers, copy from others, or collaborate in any way. Any collaborative behaviour during the examinations will result in failure of the exam, and may lead to failure of the course and University disciplinary action.

Course Procedure

The classes will be conducted through various activities including discussion of concepts and problem-solving, student initiative and active involvement, as well as practice of quantitative problems. Students are expected to actively participate and take initiative for their own learning. The final grade will be based on quizzes, an assignment, a midterm, and a comprehensive final examination.

Marks Distribution

Component	Weight	Attendance (%)	Marks
Attendance	5%	90% and above	5
Quiz	20%	85% to less than 90%	4
Assignment	10%	80% to less than 85%	3
Midterm	20%	75% to less than 80%	2
Final	45%	70% to less than 75%	1
Total	100%	Less than 70%	0

Administrative Information and Course Requirements

- Four quizzes will be given; the best three will be counted. There will be no make-up quizzes.
- There will be no make-up midterm, unless a student submits an application through the corresponding Chair of the Department at least 7 days before the midterm's scheduled date.
- No student will be allowed to sit for the final examination if he/she does not attend at least 70% of the total classes.
- Students are strongly encouraged to visit the instructor's office to ask questions and discuss problems. If unable to meet during consultation hours, appointments can be arranged at other times.

Learning Outcomes

Upon successful completion of this course, students should be able to:

- Solve systems of linear equations using multiple methods, including Gaussian elimination and matrix inversion.
- Carry out matrix operations, including inverses and determinants.
- Demonstrate understanding of the concepts of vector space and subspace, linear independence, span, and basis.
- Determine eigenvalues and eigenvectors and solve eigenvalue problems.
- Apply principles of matrix algebra to linear transformations.
- Find the Fourier series of periodic functions.

- Find the Fourier sine and cosine series for functions defined on an interval.

Grading System

Grades will be calculated using the BRAC University grading formula as follows:

Score (%)	Grade	Remark
90 – 100	A (4.0)	Excellent
85 – <90	A– (3.7)	
80 – <85	B+ (3.3)	
75 – <80	B (3.0)	
70 – <75	B– (2.7)	
65 – <70	C+ (2.3)	Fair
60 – <65	C (2.0)	
57 – <60	C– (1.7)	
55 – <57	D+ (1.3)	Poor
52 – <55	D (1.0)	
50 – <52	D– (0.7)	
<50	F (0.0)	Failure

Lecture Delivery Plan

Lecture No.	Topics
1	Matrices, matrix operations, related theorems.
2	Continuation of Lecture 1; determinants.
3	Matrix inverses and applications; method of finding inverse matrix.
4	Continuation of Lecture 3; systems of linear equations.
5	Gaussian elimination method and applications.
6	Continuation of Lecture 5.
7	Vector spaces and subspaces and applications. Quiz #01
8–9	Linear combination, linear independence and dependence, vector span and applications; basis and dimension; rank and nullity.
10–11	Linear transformations; kernel and nullity of a transformation. Quiz #02
12	Eigenvalues, eigenvectors, and diagonalization.
13	Continuation of Lecture 12.
14	Midterm Examination (tentative)
15	Introduction to Fourier series.
16	Definition of half-range Fourier series; related theorems.
17	Finding Fourier series of functions on a specified interval.
18	Continuation of Lecture 17.
19	Fourier integrals and Fourier transforms.
20	Finding Fourier transforms of some functions.
21	Fourier sine transform and cosine transform. Quiz #03
22	Problems related to Fourier transforms.
23	Continuation of Lecture 22. Quiz #04

Note: Lecture plans may be adjusted, if necessary, according to the progress of the students.



Course Outline
Department of Mathematics and Natural Sciences
Brac University

MAT 215: MATHEMATICS III: COMPLEX VARIABLES & LAPLACE TRANSFORMATIONS
Summer-2024

A. Course General Information:

Course Code :	MAT 215
Course Title :	MATHEMATICS III: COMPLEX VARIABLES & LAPLACE TRANSFORMATIONS
Credit Hours :	3
Contact Hours/Week :	3
Category	Offered by MNS for other programs
Prerequisites :	MAT 216
Co-requisites:	None

B. Course Catalog Description (Content):

Complex Variables: Complex number systems. General functions of a complex variable. Limits and continuity of a function of complex variables and related theorems. Complex differentiation and Cauchy-Riemann equations. Cauchy's integral theorem. Cauchy's integral formula. Liouville's theorem. Taylor's and Laurent's theorem. Singular points. Residue. Cauchy's residue theorem. Laplace transforms of some elementary functions. Inverse Laplace transforms. Laplace transforms of derivatives. The unit step function. Some special theorems on Laplace transforms. Solutions of differential equations by Laplace transformations. Discrete forecasting models, continuous optimization, linear regression, dimensionality reduction with PCA analysis and support vector machines.

C. Rationale of the Course: The aim of this course is to provide the student an understanding of the mathematical ideas that are required to study machine learning and signal processing.

D. **Course Objective:** The objective of this course is:

- To provide a complete grasp of complex numbers

- To give a solid background on Laplace transforms
- To introduce mathematical foundations of basic machine learning & Signal processing

E. Course Outcomes (COs):

Upon successful completion of this course, students will be able to

SL.	CO Description
CO1	Understand complex numbers, complex variables
CO2	Explore Complex differentiation and Cauchy-Riemann equations
CO3	Analyze solutions using Laplace transforms
CO4	Use mathematics for machine learning & Signal processing

F. Mapping of Course Outcomes (COs) with Program Outcomes (POs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	√					√		
CO2	√					√		
CO3	√					√		
CO4	√	√	√		√	√		

G. Course plan specifying content, COs, co-curricular activities (if any), teaching learning and assessment strategy mapped with COs:

Class Week	Topic	Teaching-Learning Strategy	Assessment Strategy	Corresponding COs
1	Complex Variables: Complex number systems. General functions of a complex variable.	Class lecture, Class discussion, Consultations	Assignment, Quiz, Examinations	CO1
2, 3, and 4	Limits and continuity of a function of complex variables and related theorems. Complex differentiation and Cauchy-Riemann equations, Analytic functions, Harmonic functions.	Class lecture, Class discussion, Consultations	Assignment, Quiz, Examinations	CO2
4, 5, and 6	Line integral of a complex function. Cauchy's integral theorem. Cauchy's integral formula. Liouville's theorem. Taylor's and Laurent's theorem. Singular points. Residue. Cauchy's residue theorem.	Class lecture, Class discussion, Consultations	Assignment, Quiz, Examinations	CO3
7, 8, and 9	Laplace transforms of some elementary functions. Inverse Laplace transforms. Laplace transforms of derivatives. The unit step function. Some special theorems on Laplace transforms.	Class lecture, Class discussion, Consultations	Assignment, Quiz, Examinations	CO3
10	Solutions of differential equations by Laplace transformations.	Class lecture, Class discussion, Consultations	Assignment, Quiz, Examinations	CO3
11	Continuous optimization and Linear regression. support vector machines	Class lecture, Class discussion, Consultations	Assignment, Quiz, Examinations	CO4

12	support vector machines. Machine Learning based problem solving	Class lecture, Class discussion, Consultations	Assignment, Quiz, Examinations	CO4
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H. Learning Materials:

Required Textbook: Complex Variables and Applications: James Brown and Ruel Churchill;
Mathematics for Machine Learning: Marc Deisenroth, Aldo Faisal, Cheng Soon Ong.

Recommended Readings: Complex Variables: A Physical Approach with Applications and
MATLAB: Steven G. Krantz

Supplementary Readings: Complex Variables (2nd edition): Schaum's Outline Series

I. Delivery method: It will mainly involve two lectures of 1 hour and 20 minutes durations every week. Different teaching aids e.g. powerpoint presentation will be used when it will be deemed necessary. Classroom discussion and problem solving sessions will supplement the lecture presentations.

J. Assessment and Evaluation:

Assessment Strategy

Evaluation of the students will be done on the basis of the marks distribution scheme described earlier. The final examination will be taken on the entire syllabus. The mid-semester examination will be one hour long and will be held around the middle of the semester. The quizzes are based on the lectures and there will be no make-up for missed quizzes. During quizzes, mid-term and final exam each student must work alone.

Marks distribution: The total mark for this course is 100. The marks will be allocated as follows:

Attendance	Quiz	Assignments	Midterm	Final
5%	20%	20%	20%	35%

Attendances policy: Arriving 15 minutes late or more is automatically absent. If the total attendance is less than 70% of the total number of classes, the student will not be allowed to appear in the mid-semester and semester final exams irrespective of the reason of absences, and he/she will be listed as an absentee. 5% of the total marks allocated for class attendance will be awarded according to the following scheme.

Attendance	Marks
90% and above	5
85% - 89%	4
80% - 84%	3
75% - 79%	2
70% - 74%	1
Less than 70%	0(Absentee)

Make-up Procedures: Make-up examinations will only be allowed to the students with extreme medical conditions OR death in the immediate family during the semester”. In such cases, permission to sit for the make-up exam is to be recommended by the course teacher and approved by the departmental head offering the course. The candidate must pay a make-up exam fee as per the university rule.

K. Grading policy: The grades and grade points will be calculated as per the Brac University grading system which is as follows:

% of marks obtained	Letter grade	Grade sign	Grade points
97 - 100	A ⁺	plus	4.00
90 - < 97	A	plain	4.00
85 - < 90	A ⁻	minus	3.70
80 - < 85	B ⁺	plus	3.30
75 - < 80	B	plain	3.00
70 - < 75	B ⁻	minus	2.70
65 - < 70	C ⁺	plus	2.30
60 - < 65	C	plain	2.00
57 - < 60	C ⁻	minus	1.70

55 - < 57	D+	plus	1.30
52 - <55	D	plain	1.00
50 - <52	D-	minus	0.70
< 50	F	plain	0.00

L. Course Coordinator:

Course Outline

STA201: Elements of Statistics and Probability

Semester : Fall 2021
 Course Title : Elements of Statistics and Probability
 Course Code : STA201
 Course Credits : 03

Course Instructors:

Initials	Name	Email	Designation	Section(s)
RFI	Dr Mohammad Rafiqul Islam	mrafiq@bracu.ac.bd	Associate Professor	3, 9
MZR	Mr. Md. Mahfuzur Rahman	mahfuz.rahman@bracu.ac.bd	Lecturer	12
TSM	Mr. Mirza Md. Tanjim Shorif Mugdho	mirza.mugdho@bracu.ac.bd	Lecturer	2
MIT	Mr. Mirza Md. Tausif Shorif Snigdho	mirza.tausif@bracu.ac.bd	Lecturer	4
LFA	Ms. Lubaba Ferdous Alim	lubaba.ferdous@bracu.ac.bd	VC's Fellow	6, 8, 10, 11
SYED	Mr. Shehran Syed	shehran.syed@bracu.ac.bd	VC's Fellow	5, 7, 13, 14
MQH	Ms. Mahfuza Haque Mahi	ext.mahfuza.haque@bracu.ac.bd	Adjunct Faculty	1

Course Coordinators:

Mr. Mirza Md. Tausif Shorif Snigdho (MIT)
 Lecturer
 Department of CSE
 School of Data and Sciences
 Brac University, Bangladesh
 Email: mirza.tausif@bracu.ac.bd

Mr. Shehran Syed (SYED)
 Lecturer, Vice Chancellor's Fellowship
 Department of MNS
 School of Data and Sciences
 Brac University, Bangladesh
 Email: shehran.syed@bracu.ac.bd

Course Curriculum:

Frequency distribution; mean, median, mode and other measures of central tendency; standard deviation and other measures of dispersion; measure of skewness; and box-whisker plot; correlation and regression analysis; elementary probability theory; conditional probability and Bayes' Theorem; random variables and joint probability distributions; discrete probability distributions: geometric, binomial and Poisson distribution; continuous probability distributions: normal and exponential distributions; statistical hypothesis testing: one-sample and two-sample z-test and t-test.

Course Learning Outcomes (CLOs):

On successful completion of this course, students should be able to:

CLO1: Distinguish between different types of data and describe data using tables and graphs.

CLO2: Summarize and analyse data using different summary measures.

CLO3: Calculate and interpret results from correlation and regression analysis.

CLO4: Demonstrate systematic understanding of the basic concepts of probability, Bayes' Theorem, and random variables.

CLO5: Compute probabilities and make inferences using discrete probability distributions and continuous probability distributions.

CLO6: Understand and apply the concepts of statistical hypothesis testing to draw conclusions from real-life scenarios.

Course Delivery and Performance Evaluation:

- There will be two lectures delivered per week, along with 3 hours of scheduled consultation.
- Students will be evaluated based on their performance on post-lectures assessments, assignments, quizzes, a viva-voce assessment, a midterm exam and a final exam.

Remote Learning Administrative Information:

- All lecture material (videos, assessments, notes, etc.) will be posted on buX on a weekly basis.
- The discussion forum on buX should be used to discuss ideas and issues related to the lecture topics.
- Section-wise consultation and communication will be conducted using platforms chosen by the section instructor. (e.g. Google Meet / Zoom / Microsoft Teams for consultation; Slack / Discord / Google Classroom for communication, etc.)
- Average scores of the best 3 (three) quizzes out of 4 (four) will be considered. There will be no make-up quizzes. All the quizzes will be administered in buX.
- There will be no makeup midterm examination, unless a student submits an application through the corresponding chair of the department before the midterm exam.
- A total of at-least 5 (five) assignments will be assigned throughout the semester. Average scores of all the assignments will be counted.
- Each assignment will be graded out of 30 raw points, of which 20 points are allocated to solving the problems, 7 points are allocated to on-time submission, and 3 points are allocated to attempting all problems, overall effort, and presentation. For submissions after the deadline, 1 mark will be deducted each day up-to a maximum of 7 marks (7 days).

Mark Distribution:

Class Participation (Post-Lecture Assessments)	10%
Assignment (Average of all)	20%
Quiz (Best 2 out of 3)	15%
Viva Voce Assessment	15%
Midterm Exam	20%
Final Exam	20%

NOTE: The mark distribution may change, if necessary, according to the circumstances of the term.

Grading Policy (Fall 2021):

Numerical Scores	Letter Grade	Grade Points
97 to $\leq$ 100	A+ (Exceptional)	4.0
90 to $<$ 97	A (Excellent)	4.0
85 to $<$ 90	A-	3.7
80 to $<$ 85	B+	3.3
75 to $<$ 80	B (Good)	3.0
70 to $<$ 75	B-	2.7
65 to $<$ 70	C+	2.3
60 to $<$ 65	C (Average)	2.0
57 to $<$ 60	C-	1.7
55 to $<$ 57	D+	1.3
52 to $<$ 55	D (Poor)	1.0
50 to $<$ 52	D-	0.7
$<$ 50	F (Failure)	0.0

Brac University Plagiarism Policy:

Cheating, Plagiarism, and Copying are serious academic misconducts, and the University has placed a “zero tolerance” approach to all unethical practices. Plagiarism is a serious **academic offence (Brac University Regulations of Students Conduct, P. 6-13)**. Students caught plagiarising may face either academic or disciplinary consequences. If found guilty of plagiarism, it may result in the following:

- A failing grade of an assignment or course
- Suspension from the university for a period of time
- Permanent expulsion from the university

Student Honour Code:

I will not cheat, use unfair means, join intentionally or unintentionally any online or offline group in which exam answers are posted or discussed, or engage in any behaviour that would commonly be deemed to be academically unethical. I will immediately notify the course teacher of any such activity or online or offline groups that I become aware of. I acknowledge that I may be suspended or expelled from Brac University if I am found to have engaged in any academically unethical behaviour.

Text References:

1. Probability and Statistics for Engineering and the Sciences - Jay Devore
2. Statistical Techniques in Business and Economics - Lind, Marchal, & Wathen
3. Introduction to Probability Models - Sheldon M. Ross

Lecture Plan:

Week	Lecture	Topics	Quiz	Asgmt.	Text: Chapter
1	1	Introduction and Basic Concepts of Statistics	1	1	1: 1
	2	Graphical Representation of Data			1: 1
2	3	Arithmetic Mean; Median; Mode		2	1: 1
	4	Geometric Mean; Harmonic Mean; Weighted Mean; Quantiles			1: 1
3	5	Variance and Standard Deviation			1: 1
	6	Empirical Rule of Standard Deviation; Shape Characteristics of Distributions			1: 1
4	7	Correlation Analysis	2	3	1: 12; 2: 13
	8	Rank-Order Correlation			2: 16
5	9	Simple Linear Regression Analysis			1: 12, 2: 13
	10	Introduction to Multiple and Logistics Regression			2: 14
6	11	Set Theory and Basic Probability	3	4	1: 2
	12	Bayes’ Theorem			1: 2
7	MIDTERM EXAMINATION				
8	13	Introduction to Random Variables and Mathematical Expectation	3	5	1: 3, 4
	14	Joint Probability and Conditioning on Random Variables			1: 5
9	15	Binomial and Geometric Distribution	4		1: 3
	16	Poisson Distribution			1: 3
10	17	Normal Distribution			1: 4
	18	Exponential Distribution			1: 4
11	19	Introduction to Statistical Hypothesis Testing		6	1: 8
	20	Terminologies of Hypothesis Testing			1: 8
12	21	One-Sample z-Test for a Population Mean			1: 8
	22	One-Sample t-Test for a Population Mean			1: 8
13	23	Two-Sample z-Test for Population Means			1: 9
	24	Two-Sample t-Test for Population Means			1: 9
	FINAL EXAMINATION				

NOTE: There may be changes in the plan, if necessary, according to the circumstances of the term.

Discrete Mathematics

A. Course General Information:

Course Code:	CSE230
Course Title:	Discrete Mathematics
Credit Hours:	3
Contact Hours:	3
Category:	Core
Prerequisites:	-

B. Course Catalog Description (Content):

Basic Counting Techniques, Rule of Sum, Rule of Product, Rule of Subtraction, Rule of Division, The Pigeonhole Principal, Permutations and Combinations, Binomial Coefficient and Identities, Discrete Probability, Probabilistic Reasoning, Probability Theory, Conditional Probability, Independence, Random Variables, Bayes' Theorem, Distributions, Expectation and Variance, Logic and Proofs, Propositional Logic, Logic Puzzles, Logic Circuits, Propositional Equivalences, Satisfiability, Predicates and Quantifiers, Sets, Set Operations, Functions, Sequences and Summations, Number Theory, Cryptography, Mathematical Induction, Recurrence Relations, Applications of Recurrence Relations, Graph and Graph Models, Terminologies of Graph.

C. Course Objective:

The objectives of this course are to

01. To develop students' mathematical maturity, the ability to understand and create mathematical arguments.
02. To prepare them for many foundational courses of CS like Data Structures, Algorithms, Automata Theory, Formal Languages, Compiler Theory etc.

D. Lesson Plan:

Week 1	Lecture 1: Introduction to Discrete Mathematics, Basic Counting Techniques, Rule of Sum, Rule of Product, Rule of Subtraction, Rule of Division
	Lecture 2: The Pigeonhole Principal, Permutations and Combinations
Week 2	Lecture 3: Binomial Coefficient and Identities
	Lecture 4: Introduction to Discrete Probability
Week 3	Quiz 1 (Counting) and Discussion.
	Lecture 5: Probabilistic Reasoning, Probability Theory
Week 4	Lecture 6: Conditional Probability, Independence
	Lecture 7: Random Variables, Bayes' Theorem
Week 5	Lecture 8: Distributions, Expectation and Variance
	Lecture 9: Logic and Proofs, Propositional Logic, Logic Puzzles, Logic Circuits
Week 6	Quiz 2 (Discrete Probability) and Discussion.
	Lecture 10: Propositional Equivalences, Satisfiability, Predicates and Quantifiers
Week 7	<i>Midterm Week</i>
Week 8	Lecture 11: Sets, Set Operations
	Lecture 12: Functions, Sequences and Summations
Week 9	Lecture 13: Number Theory
	Quiz 3 (Sets and Functions) and Discussion.
Week 10	Lecture 14: Cryptography
	Lecture 15: Mathematical Induction
Week 11	Lecture 16: Recursive Algorithms
	Lecture 17: Application of Recurrence Relations
Week 12	Quiz 4 (Number Theory) and Discussion.
	Lecture 18: Graph and Graph Models (Optional) or Review

Course Title: Numerical Methods

Course # CSE 330. All Sections

Fall Semester 2021

Department of Computer Science & Engineering

46 Mohakhali, Dhaka - 1212

Course Coordinator : Abu Mohammad Khan

• Instructor Information

Section #	:	Name & Email
Sec # 1	:	Rafeed Rahman (N03); Email - ext.rahman.rafeed@bracu.ac.bd
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Lab for CSE330	:	Sifat Ut Taki (M06); Email: ext.sifat.taki@bracu.ac.bd

• Course Information

Class Meeting Time	:	Classes are online at https://bux.bracu.ac.bd/ Watch the video lectures at your convenience followed by a short quiz on the video lecture materials
Online Live sessions	:	Sec#1: 11:00am -12:20pm on MW each week. Sec#2: 8:00am - 9:20am on MW each week. Sec#3: 2:00pm - 3:20pm on MW each week. Sec#4: 8:00am - 9:20am on ST each week Sec#5: 11:00am -12:20pm on ST each week. Sec#6: 2:00pm - 3:20pm on ST each week. Sec#7: 3:30pm - 4:50pm on ST each week at https://meet.google.com/bng-jqdf-ikc Sec#8: 3:30pm - 4:50pm on MW each week. Sec#9: 12:30pm - 1:50pm on ST each week at https://meet.google.com/waf-xvpg-kxq Sec#10: 12:30pm - 1:50pm on MW each week
Required Text	:	Numerical Analysis II – Lecture Notes. Author(s): Anthony Yeates.
Reference Book	:	Numerical Methods In Engineering With Python. Author: Jaan Kiusalaas.
Course Objectives	:	The course will help students to recognise the need for numerical analysis, and importance of error analysis. They will learn various methods to linearize a polynomial, differentiate and integrate different functions by using approximations, and finally how to solve the linearized equation by using the laws of linear algebra, like Gaussian elimination, QR decomposition, etc. We will also discuss about the numerical Integration methods.
Grade Distribution	:	The final grade is based on the Graded Quizzes, Assignments, lab, midterm, final exam and Course viva contributing 10%, 20%, 20%, 20%, 20% and 10% respectively. Since there is a high possibility of multiple tests, quizzes, etc. we offer you an option to choose any seven (07) quizzes out of 10 and any seven Assignments out of 10 for your final grades. This option will help you to overcome the difficulty due to multiples quizzes, on the same day or overlapping quizzes. The format of the midterm and final exams will be announced later

• Rules and Regulations

Attendances policy	:	There is no attendance for online classes. But it is very essential need to watch the lecture videos each week regularly to keep up with the pace of the online course.
Dishonesty policy	:	During quizzes, mid-term and final exams each student must work alone. Any kind of unauthorized contribution(s) will be treated as cheating and will be given a zero. But you are free to use any software and/or advanced calculator.

- Course Objectives and the Learning Outcome:

After successful completion of the course, a student will achieve certain skills, and these skills are classified as the Course Learning Outcome (CLO)s according to Bloom's Taxonomy. For this course, the CLOs along with their assessment methods and tools are the following:

CLO-#	Outcome types	Bloom's Taxonomy level	Delivery method	Assessment tools
CLO-1	Demonstrate an understanding of the fundamental concept of numerical analysis and the need for error analysis.	C1, C2, P1	Lecture, Discussion	Assignment, Quiz, Exam
CLO-2	Interpolate polynomial functions, test their convergence, examine interpolation error	C3, C4, P1	Lecture, Discussion	Assignment using python, exam, Quiz
CLO-3	Differential by finite difference method and estimate rounding error, solve numerically the nonlinear equations by various methods, like Newton's method, etc. Numerical integration by Newton-Cotes method	C4, P1	Lecture, Discussion	Assignment, Quiz lab, exam
CLO-4	Solve linear equations by Gaussian method, QR decomposition	C3, C4	Lecture, Discussion	Quiz, lab exam, assignment,

- Lecture Details: The tentative video/lecture schedule and the exam dates are given below. The lectures are divided into ten modules based on the topics to be covered, and each module contain several lectures. The chapter numbers are according to Yeats Notes. The midterm and final exam dates are fixed by the authority (see academic calender)

Date	: Topics
Module # 1	: Floating Point Arithmetics (Ch: 1). Pop Quizzes. Graded Quiz #1 (18/10/2021) and Assignment # 1 due 19/10/2021
Module # 2	: Polynomial Interpolation - (Ch: 2.0-2.2). Pop Quizzes. Graded Quiz #2 (25/10/2021) and Assignment # 2 due 26/10/2021
Module # 3	: Polynomial Interpolation - (Ch: 2.3-2.5). Pop Quizzes. Graded Quiz #3 (01/11/2021) and Assignment # 3 due 02/11/2021
Module # 4	: Polynomial Interpolation - (Ch: 2.6-2.7). Pop Quizzes. Graded Quiz #4 (08/11/2021) and Assignment # 4 due 09/11/2021
Module # 5	: Differentiation - (Ch: 3.0-3.3). Pop Quizzes. Graded Quiz #5 (15/11/2021) and Assignment # 5 due 16/11/2021
Midterm Week	: The midterm exam will be held on November 24, 2021 from 5:00pm-6:00pm .
Module # 6	: Nonlinear equations (Ch: 4.0-4.2). Pop Quizzes. Graded Quiz #6 (29/11/2021) and Assignment # 6 due 30/11/2021
Module # 7	: Nonlinear equations (Ch: 4.3-4.4, 4.6-4.7). Pop Quizzes. Graded Quiz #7 (06/12/2021) and Assignment # 7 due 07/12/2021
Module # 8	: Linear Equations (Ch: 5.0-5.4). Pop Quizzes. Graded Quiz #8 (13/12/2021 and Assignment # 8 due 14/12/2021
Module # 9	: Least-Squares Approximatopns (Ch: 6.0-6.3). Pop Quizzes. Graded Quiz #9 (20/12/2021) and Assignment # 9 due 21/12/2021
Module # 10	: Numerical Integration (Ch: 7.0-7.3). Pop Quizzes. Graded Quiz #10 (27/12/2021) and Assignment # 10 due 28/12/2021
Final Exam Week	: The final exam will be held on January 12, 2022 (Saturday) from 5:00pm-6:30pm .